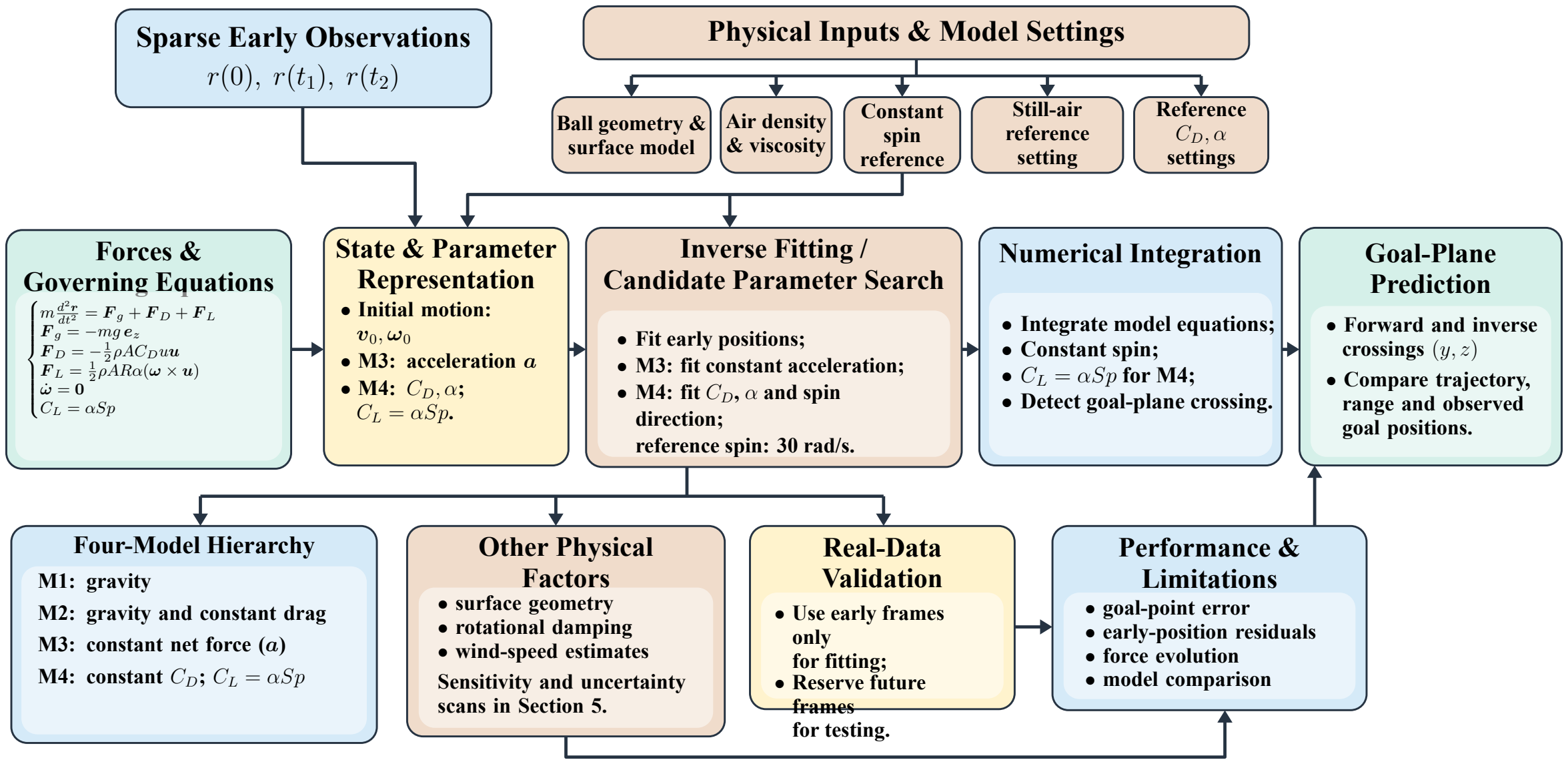
Sparse Early Observations
$r(0),\ r(t_1),\ r(t_2)$
Physical Inputs & Model Settings
Ball geometry & surface model
Air density & viscosity
Constant spin reference
Still-air reference setting
Reference C_D, α settings
Forces & Governing Equations
$m\frac{d^2\boldsymbol{r}}{dt^2} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L$
$\boldsymbol{F}_g = -mg\,\boldsymbol{e}_z$
$\boldsymbol{F}_D = -\frac{1}{2}\rho A C_D u \boldsymbol{u}$
$\boldsymbol{F}_L = \frac{1}{2}\rho A R \alpha(\boldsymbol{\omega} \times \boldsymbol{u})$
$\dot{\boldsymbol{\omega}} = \mathbf{0}$
$C_L = \alpha S p$
State & Parameter Representation
• Initial motion: $\boldsymbol{v}_0, \boldsymbol{\omega}_0$
• M3: acceleration $\boldsymbol{a}$
• M4: C_D, α; $C_L = \alpha S p$.
Inverse Fitting / Candidate Parameter Search
• Fit early positions;
• M3: fit constant acceleration;
• M4: fit C_D, α and spin direction;
reference spin: 30 rad/s.
Numerical Integration
• Integrate model equations;
• Constant spin;
• $C_L = \alpha S p$ for M4;
• Detect goal-plane crossing.
Goal-Plane Prediction
• Forward and inverse crossings (y, z)
• Compare trajectory, range and observed goal positions.
Four-Model Hierarchy
M1: gravity
M2: gravity and constant drag
M3: constant net force ($\boldsymbol{a}$)
M4: constant C_D; $C_L = \alpha S p$
Other Physical Factors
• surface geometry
• rotational damping
• wind-speed estimates
Sensitivity and uncertainty scans in Section 5.
Real-Data Validation
• Use early frames only for fitting;
• Reserve future frames for testing.
Performance & Limitations
• goal-point error
• early-position residuals
• force evolution
• model comparison